

Vincent Liu

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Experience

- Exia Labs**
Software Engineer Intern

Nov 2025 - Present
Bellevue, WA - Remote

 - Develop and deploy a multi-agent RAG Chatbot platform used by the U.S. Military Academy, West Point to answer research queries on Chinese PLA theories, tactics, and doctrine.
 - Implement unit movement and sustainment features for a military operations planning platform, including vehicle management, automated fuel calculations, and route planning.
 - Design and build a clean, investor-facing website with modular components, emphasizing simple navigation and polished presentation for investors and partners.
- UXLy**
Software Engineer Intern

Jan, 2025 - May, 2025
San Francisco, CA - Remote

 - Integrate and optimize chatbot guardrails to prevent unsafe or off-topic responses, ensuring compliance with guidelines while enhancing user satisfaction and engagement.
- UC Santa Cruz, Baskin School of Engineering**
Reader

Jan 2025 - Mar, 2025
Santa Cruz, CA

 - Evaluate and provide feedback on assignments for 300+ students to determine students' understanding of Agile workflow and core software engineering principles.

Projects

- Exia Labs - Red:**
 - Multi-agent AI system for answering complex queries on Chinese PLA military branches, leveraging a RAG architecture with semantic vector search, document retrieval pipelines, and tool-assisted research workflows.
- UXLy - Multi-tool Customer Support Chatbot:**
 - AI Chatbot with tool orchestration that maintains dialogue context, generates product recommendations, and executes order management actions within an eCommerce platform.
 - Build a full-stack eCommerce platform and integrate an AI Chatbot, engineering frontend UI components, backend APIs, authentication flows, and guardrail logic.
- SecureAI: GitHub Repository Vulnerability Scanner**
 - Full-stack web tool that analyzes GitHub repositories for security vulnerabilities using OpenAI and Gemini models, providing severity visualization and AI-generated mitigation strategies.
 - Engineer backend services including a memory-efficient GitHub repository file-fetching algorithm that filters irrelevant files, integrates API calls, and performs LLM prompting for AI-generated vulnerability analysis.
- NoteSheet Editor: Exam NoteSheet Generator**
 - Full-stack application that generates optimized notesheets for students, using a backend algorithm inspired by the 2D knapsack problem to maximize whitespace efficiency and layout quality for PDF generation.
 - Lead frontend development, building core UI components including a rich text editor.

Relevant Coursework

Programming Languages	Software & Web Development	Theory / Foundations
C / C++ Programming Assembly Language and Lab Abstract Python	Full Stack Web Development I Software Design Project I + II + III Software Design Project IV (Accel.) Computer Systems Design	Data Structures and Algorithms Introduction to Algorithm Analysis Cryptography Computer Architecture

Education

University of California, Santa Cruz
Bachelor of Science - B.S. Computer Science

2021 - 2025

Skills

- Programming Languages:** Python, C++, C, TypeScript (TSX), JavaScript (JSX), SQL, Assembly
 - Frameworks:** React, FastAPI, Flask, LangChain, Node.js, Express.js, Tailwind CSS
 - Software Principles:** Agile, Scrum
- Libraries & Tools:** PyTorch, Docker, OpenAI Agents SDK, Uvicorn, ReportLab, Tavily
 - Web & DB Technologies:** REST APIs, PostgreSQL, MongoDB, Pinecone, Clerk, WebGL, HTML5

Honors, Awards, and Certificates

- Practical Introduction to Quantum-Safe Cryptography** — Issued by IBM, Feb 2025
- Information Systems Design and Management** — Issued by Information Technology Academy, Jun 2021
- State Seal of Biliteracy (Japanese)** — Issued by State of California, Jun 2021